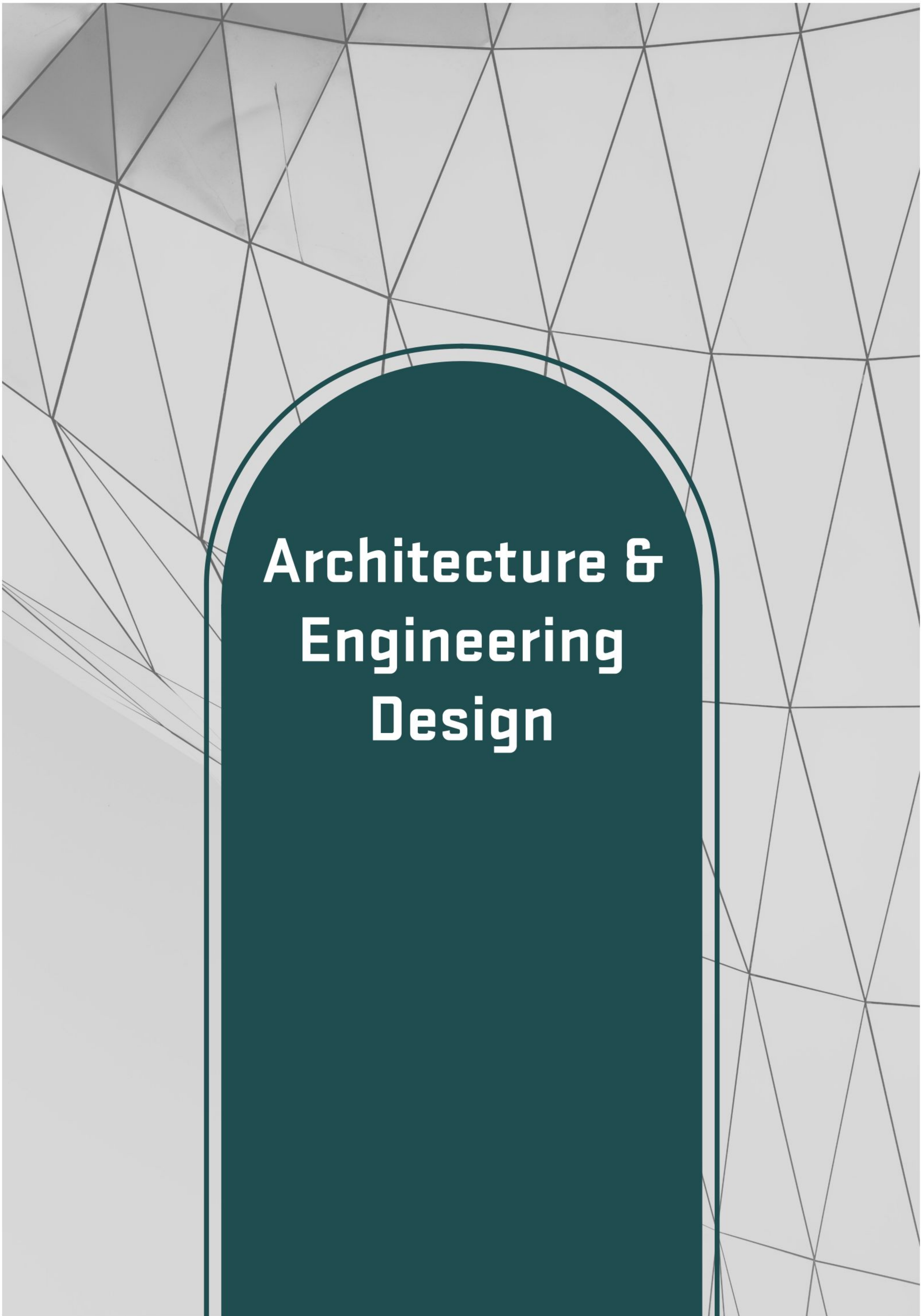
# College of Engineering & Technology

Programs

# DEPARTMENTS OVERVIEW

- Architecture & Engineering Design
- Aviation Science
- Computer Science
- Construction Technologies
- Digital Media
- Engineering Technology
- Engineering
- Information Systems & Technolgy
- Technology Management
- Transportation Technologies



# Architecture & Engineering Design

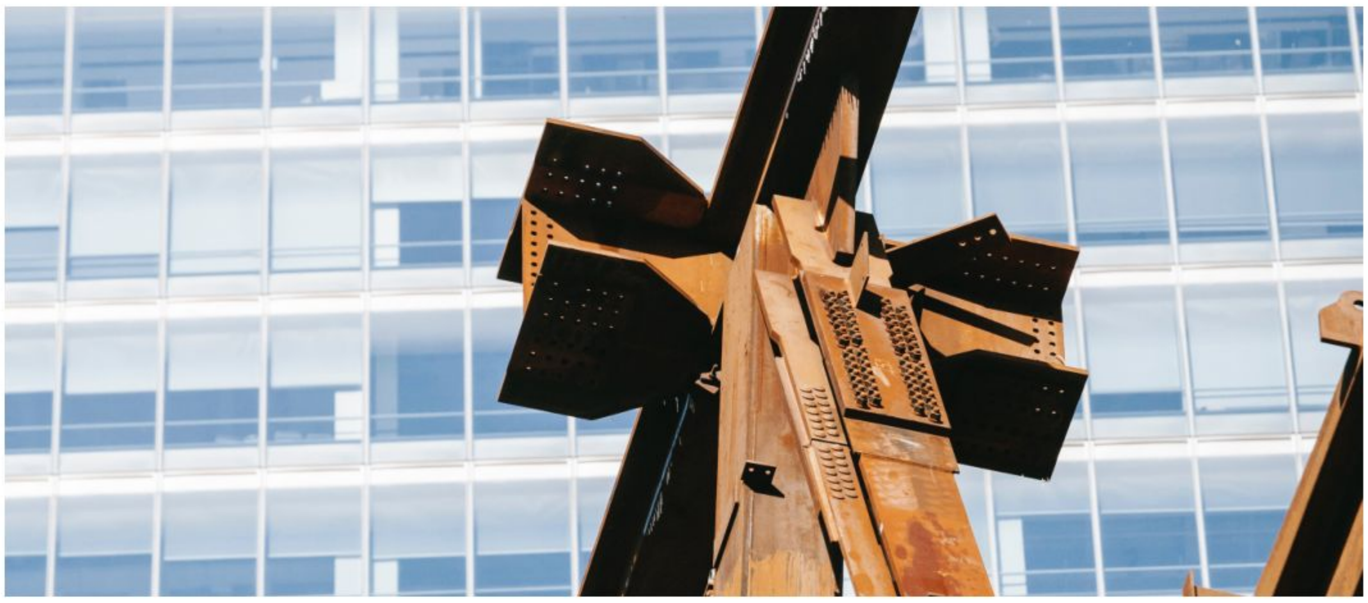
# Architecture & Engineering Design

## The Bachelor of Architecture

A 5-year professional degree. It features a rigorous design-oriented curriculum with a solid foundation in technology, practice-based coursework, plan and document generation, building codes, specifications, digital parametric modeling, building information modeling, architectural visualization, digital fabrication, building envelope systems, structural systems, and building sustainability. Students will become experts in current design and building technologies, making them ideal employees in architecture offices and related design & construction industries including civil, mechanical, and electrical.

## Engineering Design Technology

(formerly Engineering Graphics Design Technology) Works in collaboration with engineers, architects, production and industry professionals to develop new designs, products, and solutions. Students analyze and solve real-world problems using discipline relevant content and technology. Typical industrybased design resources include rough sketches, design calculations and layouts, CAD layouts, plans, elevations, details, maps, and illustrations. These same resources are used in the classroom to generate designs using 2D and 3D CAD, scale models, visual representations, and manufacturing methods. Skills learned in this area are employed in the mechanical, electrical, electronic, structural, architectural, civil, piping, technical illustration, product development, and other related fields. Degrees offered: AS and AAS (Engineering Design Technology), CP (Civil Design Technology, Mechanical Design Technology, Structural Design Technology).



# Architecture & Engineering Design

## Surveying and Mapping

Prepares students for a profession in Surveying, Mapping, and Geospatial Science on a state, regional, national and international level in public, private, and academic settings. In this program, students will be able to demonstrate knowledge and skills in data acquisition, modeling, analysis, integration, and management of geospatial reference data used to produce deliverables for land surveying, civil engineering, cartography, geographic information systems (GIS), geodesy, and remote sensing. Degree offered: AS, BS (Surveying and Mapping), AAS (Surveying Technology), CP (GIS, Surveying Technology, Civil Design and Surveying Technology)

# Aviation Science

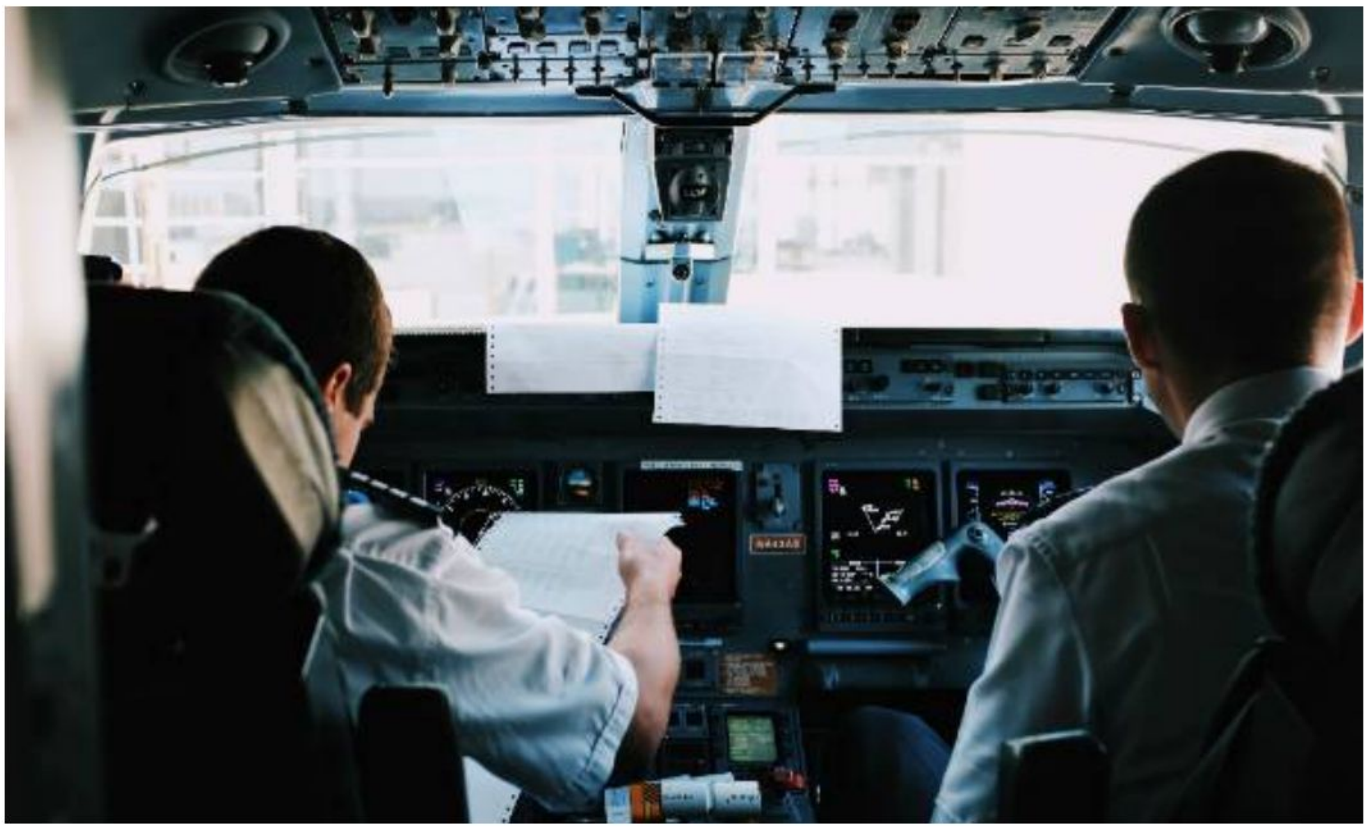
# Aviation Science

## Aviation Management

Teaches a combination of general management skills and industry knowledge, exposing students to the seemingly limitless career opportunities in the aerospace industry. The degree serves as a foundation, providing necessary knowledge, vital experiences, internships, certifications, and guidance on any additional training or experience that may be necessary for a chosen career path. Degrees offered: BS in Aviation Management.

## Professional Pilot

Prepares students for employment with airlines as well as private and corporate operators. FAA flight training is required as part of the curriculum and graduates leave with a minimum Commercial Pilot Certificate, with the option of additional certificates up through Multi-Engine Instructor that can be used as electives. UVU graduates are employed by U.S. and international airlines, cargo carriers, corporate flight departments, and many others. Degrees offered: BS in Professional Pilot.



# Computer Science

# Computer Science

## Computer Scientist

Use the theory and practice of computing to explore new and exciting ways to create software systems. They develop software in areas such as artificial intelligence, machine learning, and database for websites like Amazon. Algorithms developed by computer scientists enable us to search the web through Google and link with friends through Facebook. Degrees offered: AS, AAS, CC, BS

## Software Engineer

Design and develop large software systems. They may lead teams of software developers or quality assurance engineers. They also work with users and customers to understand their needs. Software systems, such as Microsoft Office, are implemented by software engineers. Degrees offered: BS

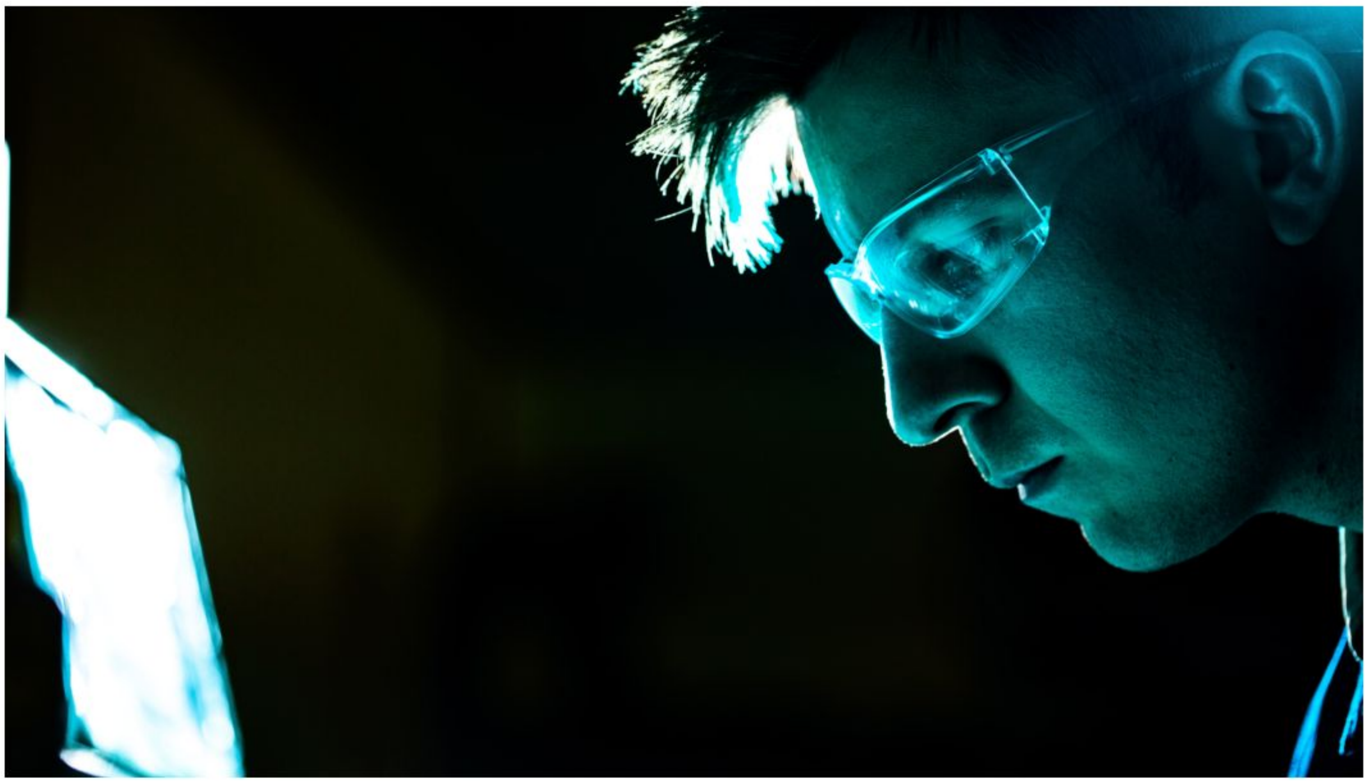
# Computer Science

## Software Developers

Implement software. They write code, participate in detailed design, and test software. They often work under the direction of software engineers and computer scientists to create software systems. All the software products you use were implemented by software developers, together with software engineers and computer scientists. Degrees offered: Bachelor of Applied Science in Software Development.

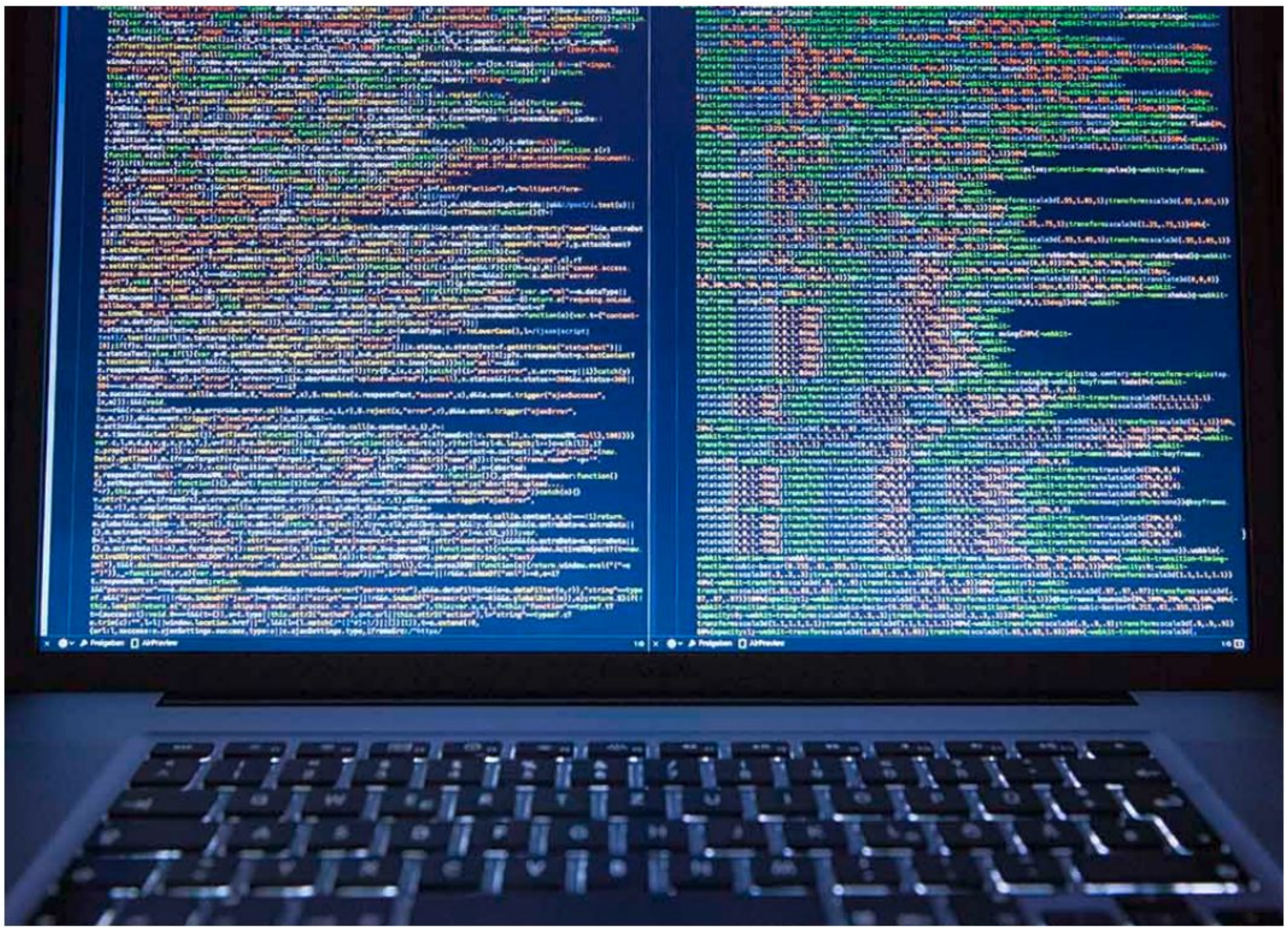
## Data Scientist

Design software to make sense of large sets of data. They use advanced techniques such as artificial intelligence, machine learning, and statistical analysis to create software used by data analysts to glean knowledge from data. Data science is one of the fastest-growing fields in all of computing. Many companies use tools developed by data scientists to learn more about their customers and serve them better. Degrees offered: Bachelor of Science in Computational Data Science.



## Master of Computer Science

focuses on preparing students to take on leadership roles and become industry innovators. They may be technical managers or software architects. They are experts in advanced topics such as deep learning and large-scale system implementation.



# Construction Technologies

# Construction Technolgies

## Construction Management

Programs include the knowledge, technical, administrative and communication skills necessary to succeed in the construction industry. Students will demonstrate the knowledge and skills to deliver construction projects with respect to scope, schedule, budget, quality, safety, and the environment. Construction Technologies include Heavy Civil, Commercial, and Residential sectors. Degrees offered: BS, AAS, CC, CP

# Facilities Management

Emphasizes the management of complex facilities. Students also learn contracts, estimating, scheduling, and CAD for such projects – allowing them to be successful in this expanding field. Degree offered: AAS

# Architecture & Engineering Design

## Cabinetry & Architectural Woodworking

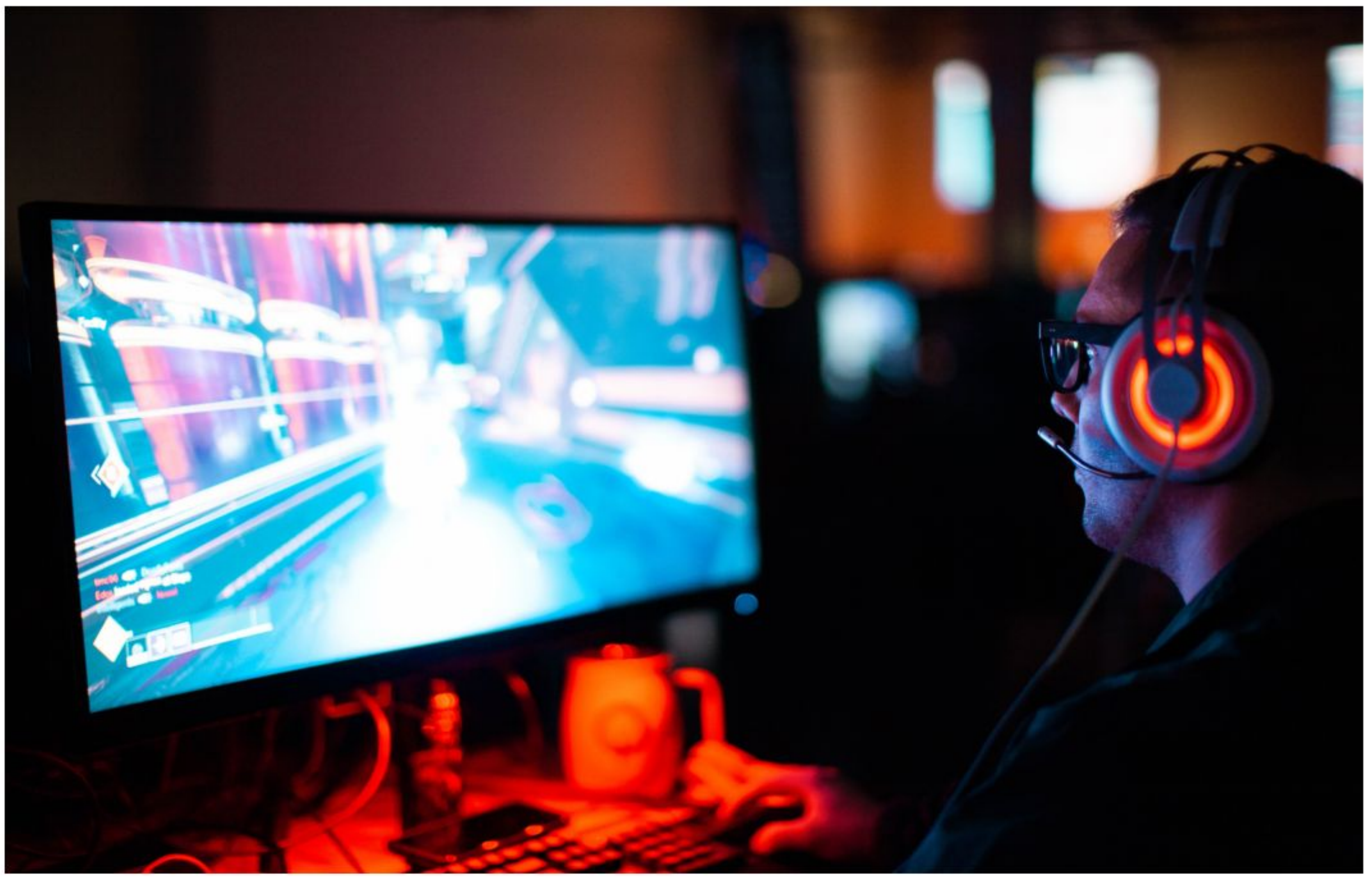
Offers students the knowledge and skills of modern woodworking, including software utilization for CNC machines and other current artistic woodworking methods. With a few extra courses, students can graduate with a High School teaching credential as well. Degrees offered: AAS, AS, CC, CP

# Digital Media

## Digital Media

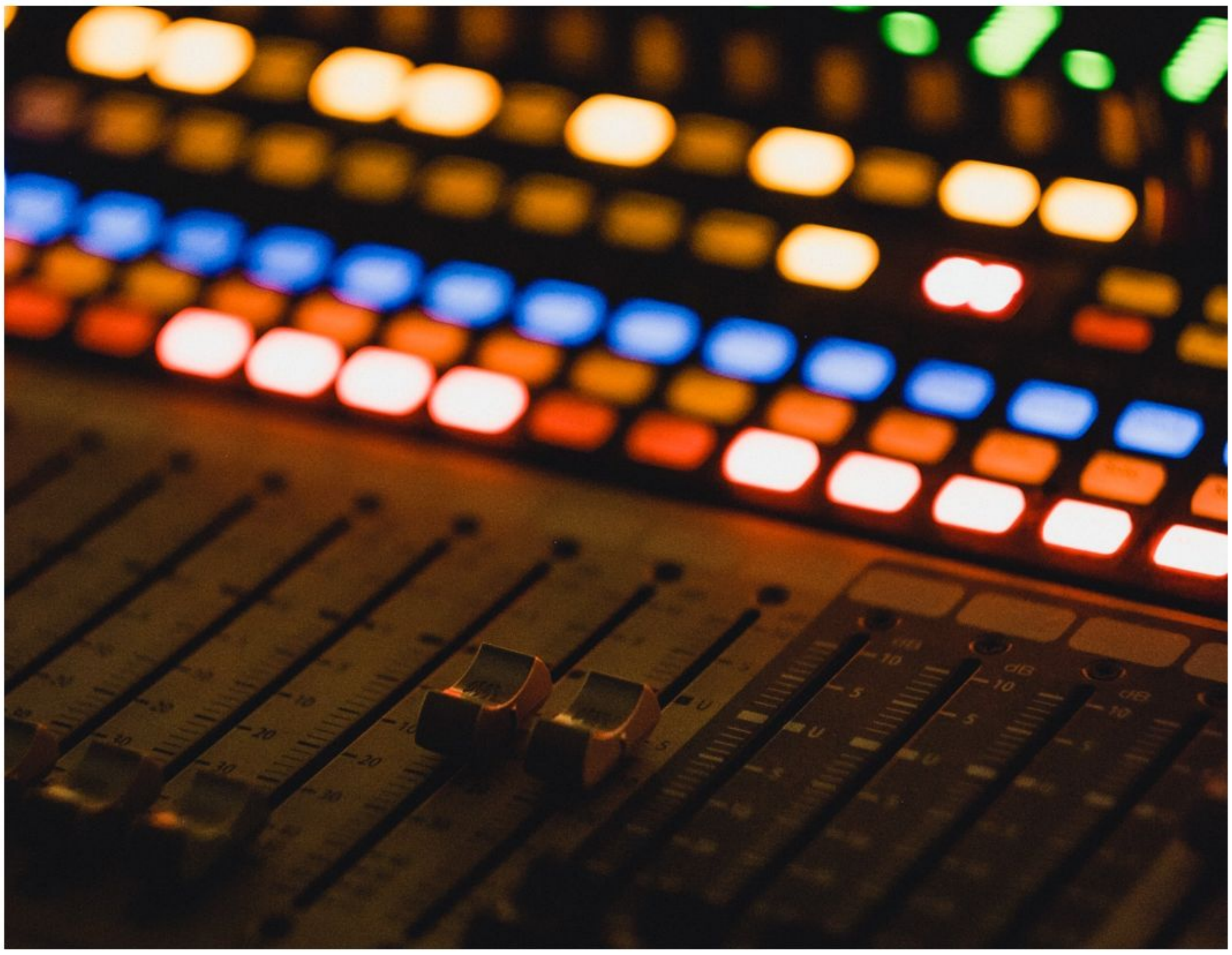
### Animation & Game Development

Provides students the opportunity to gain skills that are central to industry best practices in preparation for positions as technical artists or technical directors, with a focus on aesthetics, scripting, and 3D modeling. Degrees offered: BS (Animation and Gaming Development for 2D Animation, Animation and Gaming Development for 3D Animation/Mobile Game Development, Animation and Gaming Development for AAA/Serious Game Development)



# Digital Audio

A powerful gateway into the fascinating world of album recording and mixing, location and postproduction sound for film and video, audio restoration and forensics, live sound, radio production, audio for video games, and audio hardware and software design. Degrees offered: AAS and BS (Digital Audio Production)



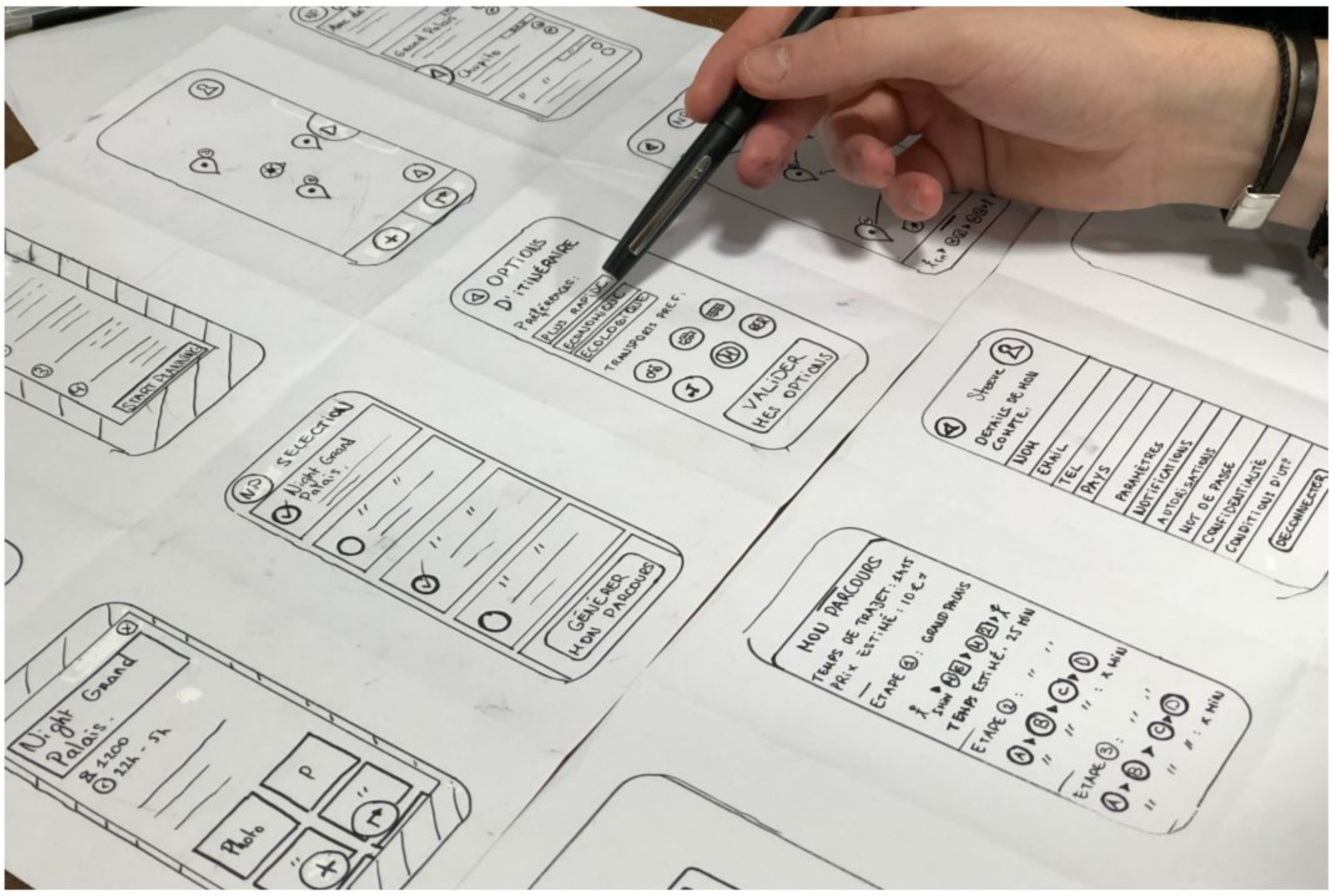
## Digital Cinema Production

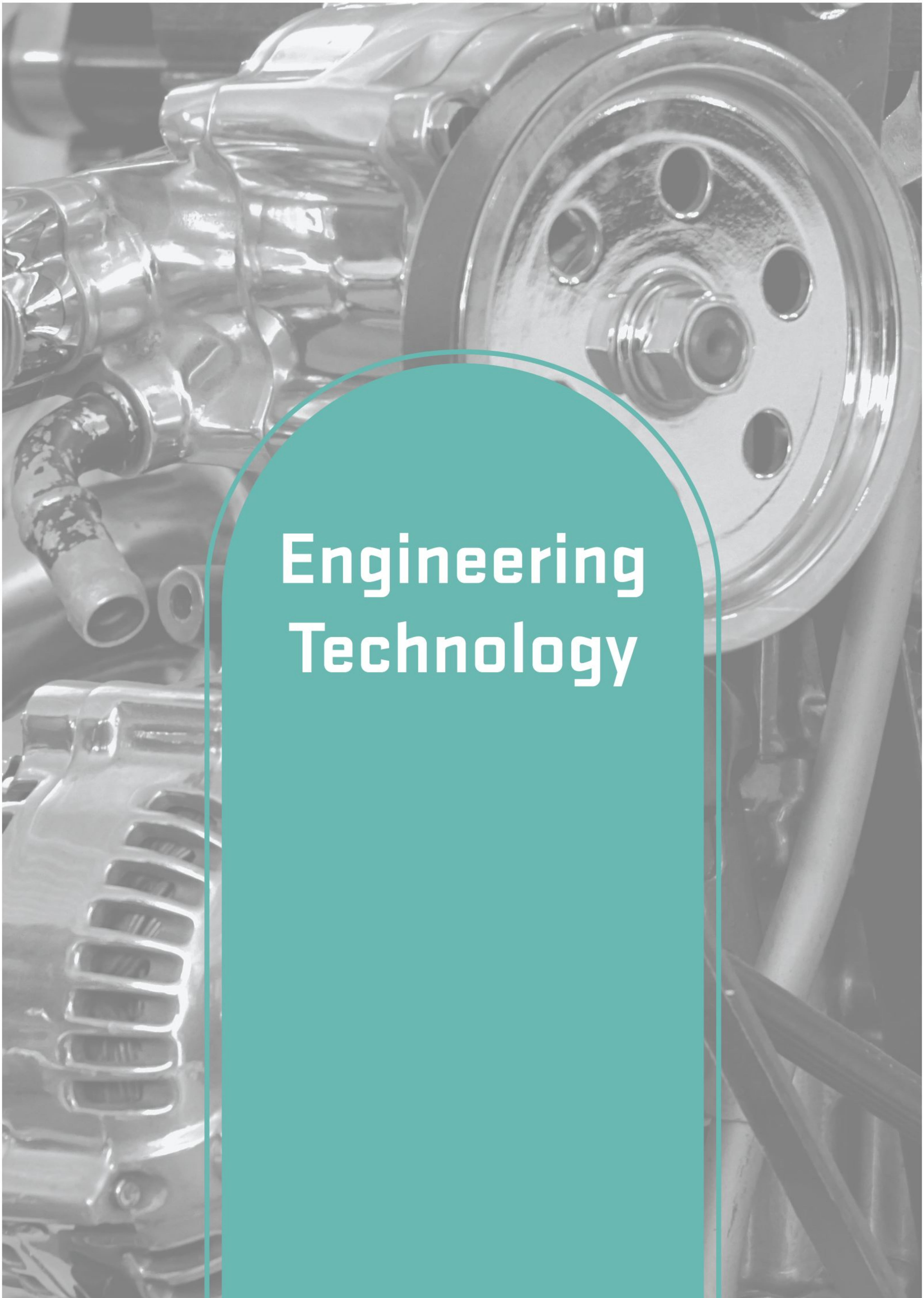
Trains students in the development, production, and post-production process of filmed media content for film, television, and web-based entertainment. Advanced students can focus their study on different skill sets including Directing for Digital Cinema, Writing for Digital Cinema, Cinematography, Production, Post-Production, Documentary, and Sports Broadcasting. Degrees offered: AAS and BS (Digital Cinema Production)

## Digital Media

### Web Design & Development

Fuses together web design, interaction design, digital product design, front-end web development, web and app development, and delivery of rich media content through the medium of the Internet to handheld mobile devices as well as desktop computers. Degrees offered: AAS (Web Design and Development), BS (Web Design and Development for Front End Development), BS (Web Design and Development for Interaction Design)



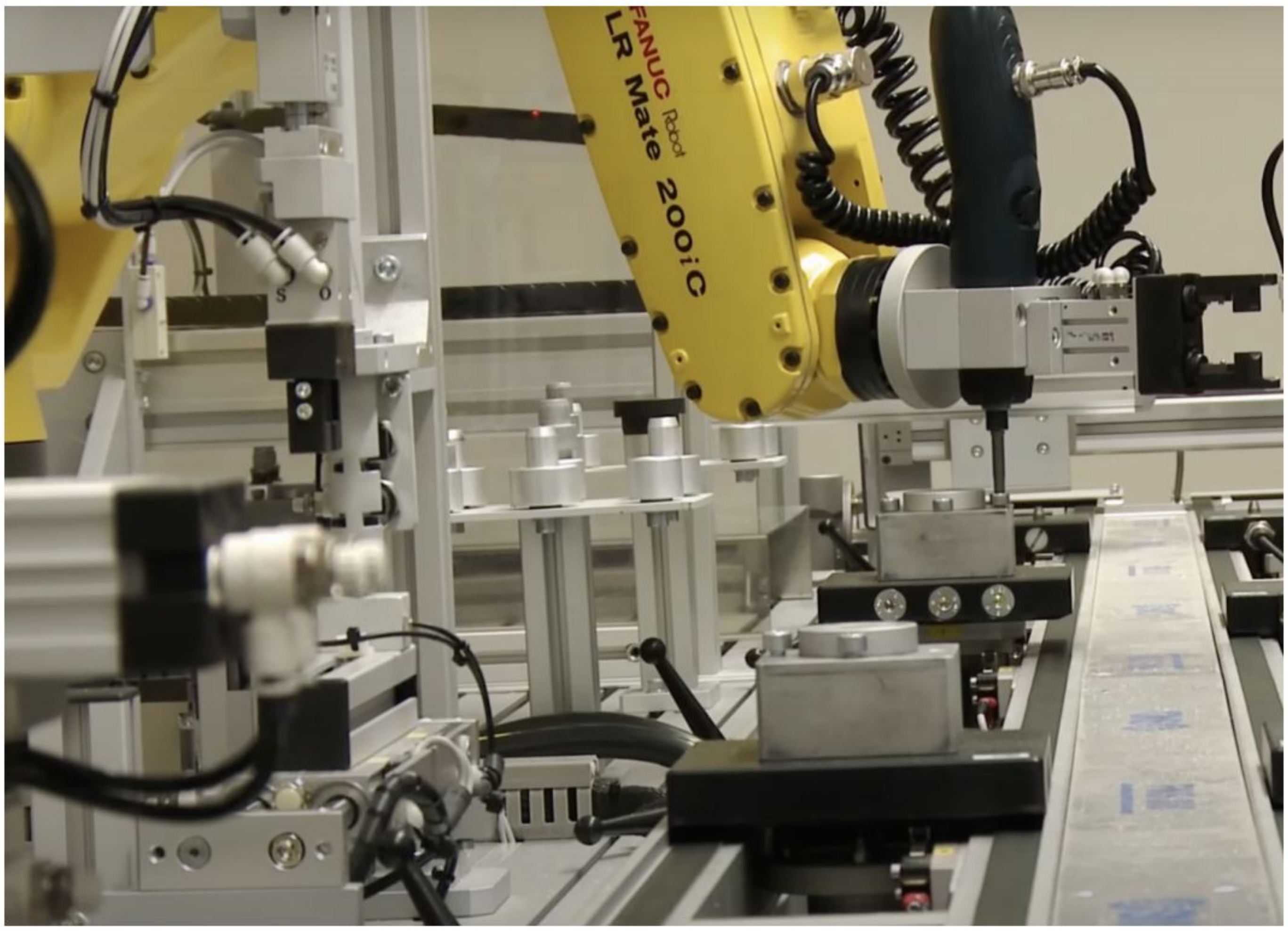


# Engineering Technology

Engineering Technology

# Automation & Electrical Technology

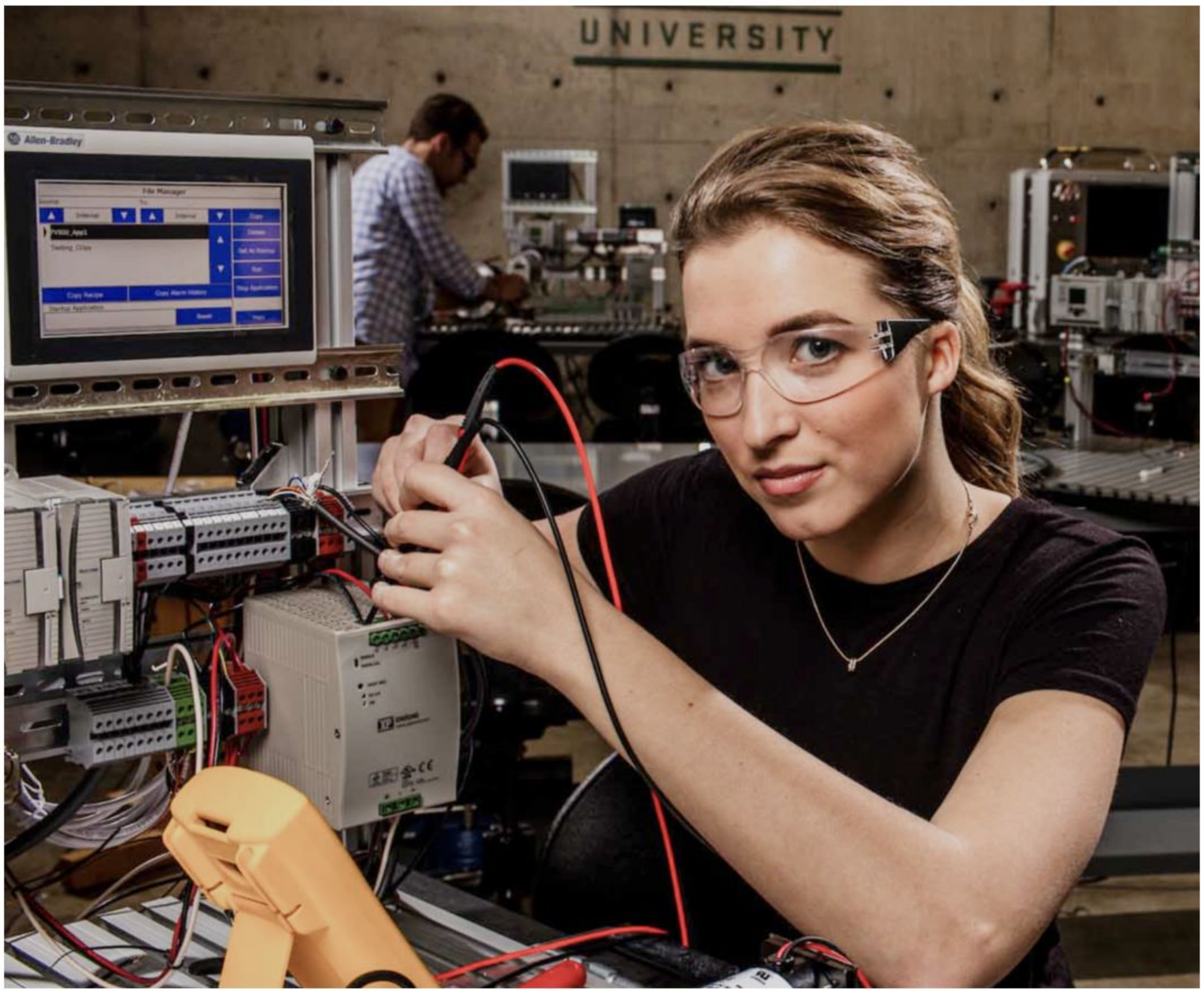
Electrical Automation technologists work with the latest automation technology that control manufacturing and industrial facilities, processes, and machines. Automation technologists troubleshoot, wire, repair, adapt, maintain, program, design, integrate, and control fully functional automated electrical systems found in a ubiquitous industry. Degrees offered: AAS, AS, CP

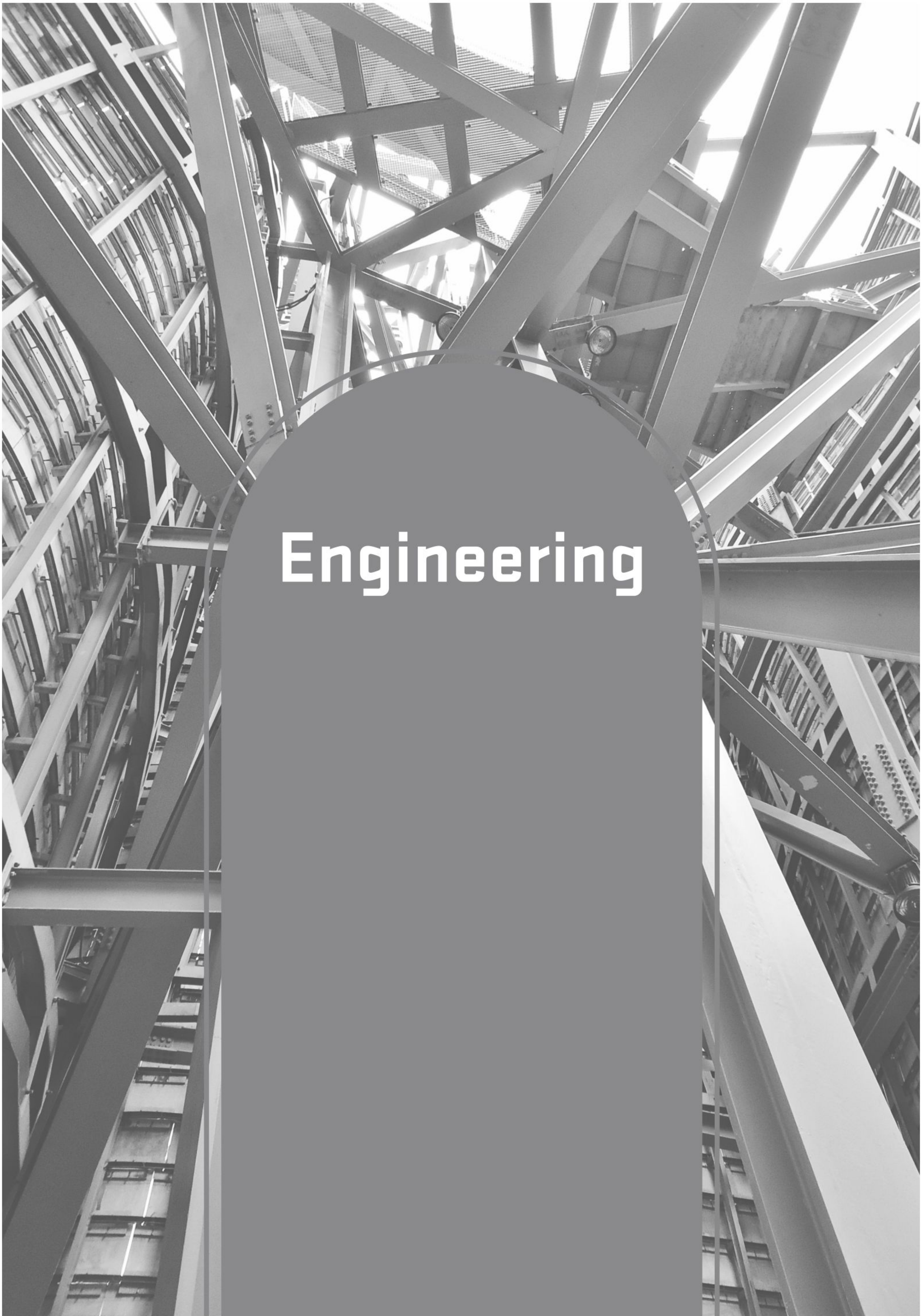


## Engineering Technology

# Mechatronics

Provides students with in-depth training in areas such as: Automation Motors, Industrial Networks, Industrial Robots, CNC Machines, and Advanced Pneumatic Designs. It is focused on designing machines and automated systems needed by industry. Degrees offered: AAS, BS

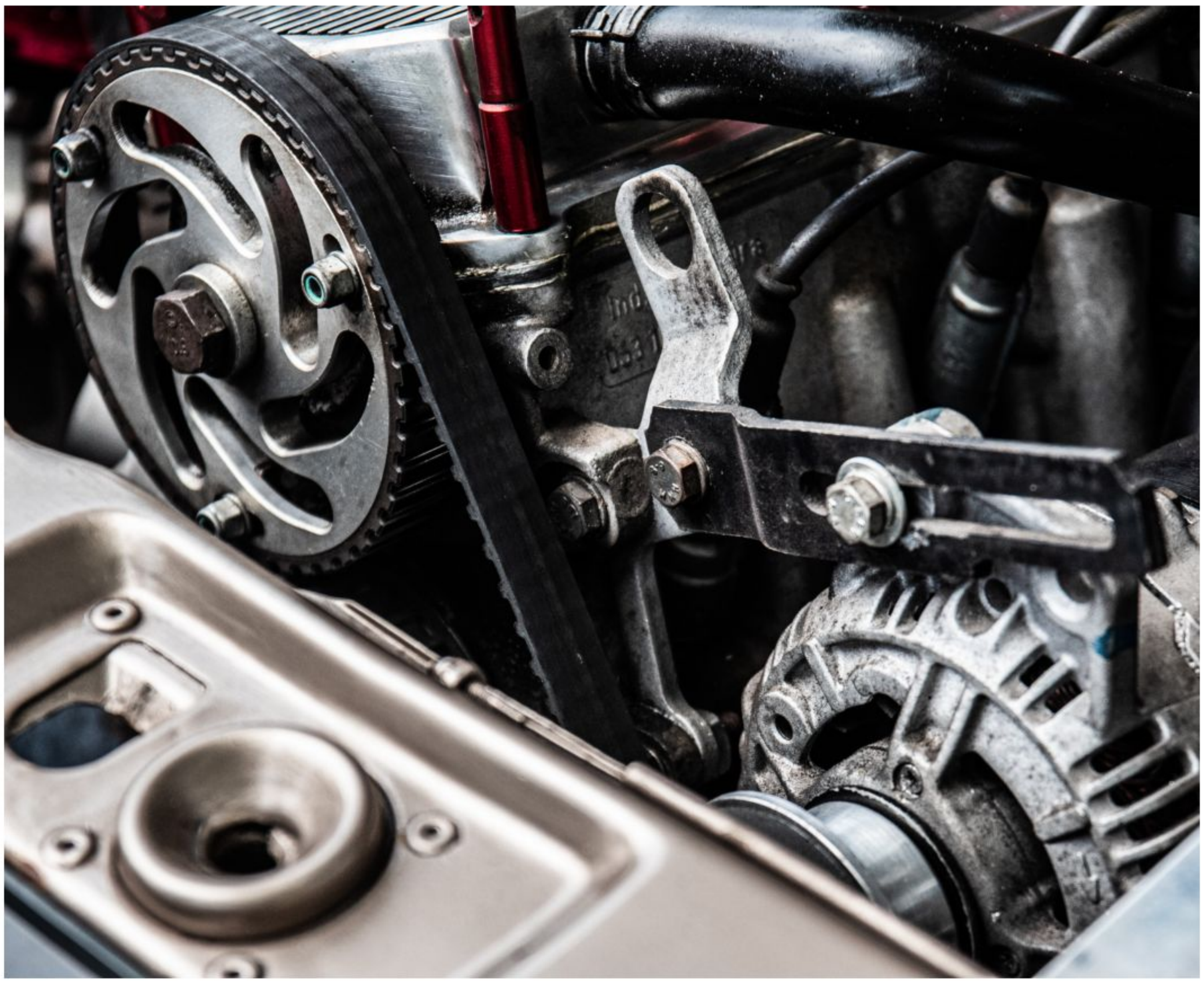




# Engineering

## Mechanical Engineering

Students are prepared to work for local, state, and federal governments; biomedical and aerospace fields; manufacturing sectors; consulting firms; transportation industry; and high tech and energy sectors. Degree offered: BS in Mechanical Engineering.

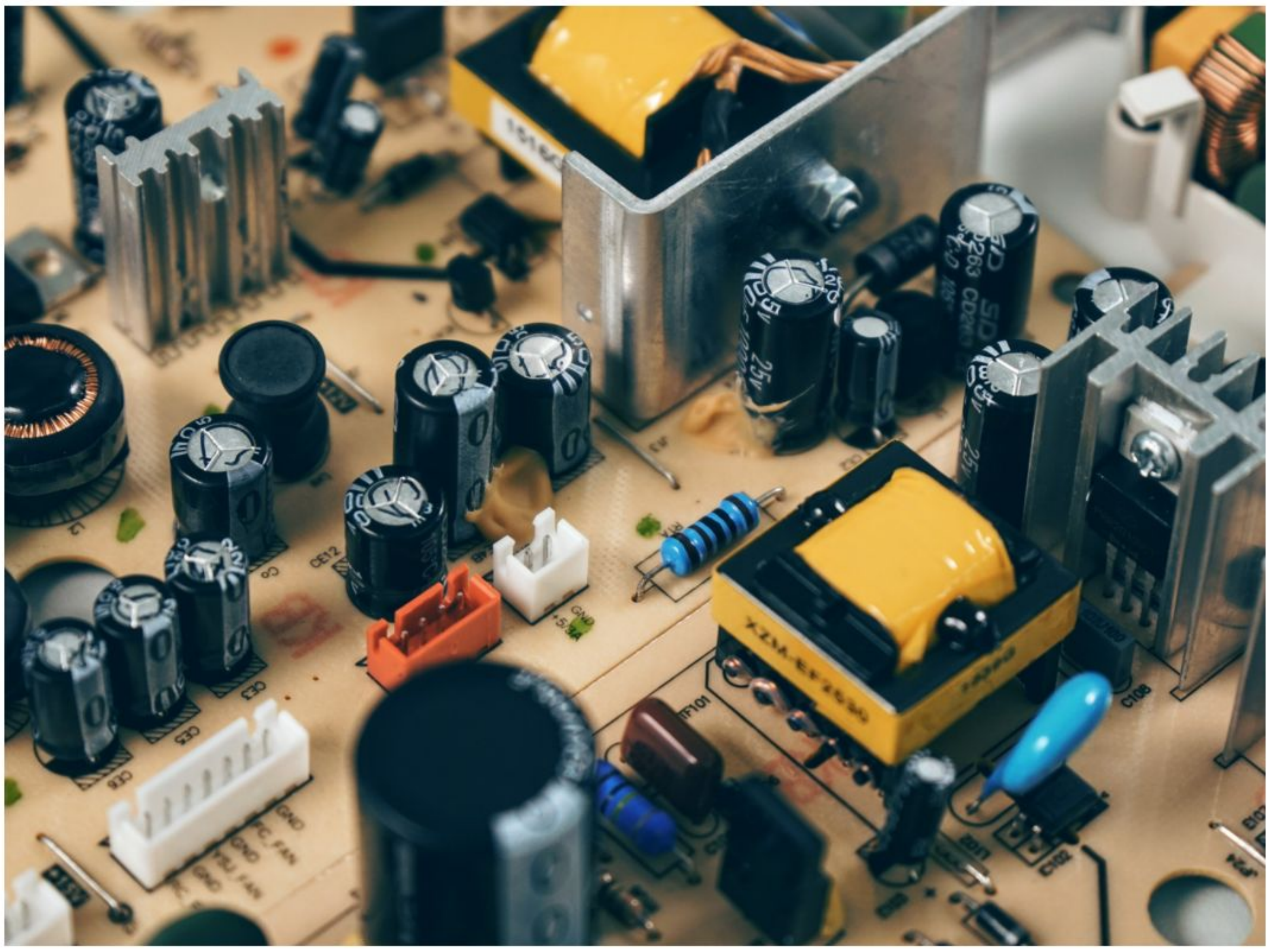


## Civil Engineer

Will apply scientific principles to the design and supervision of infrastructure components including buildings, roads, bridges, dams, tunnels, mass transit systems, and airports. Civil engineers are also involved in environmental studies and the design and supervision of municipal water supplies and sewage systems. Degree offered: BS in Civil Engineering.

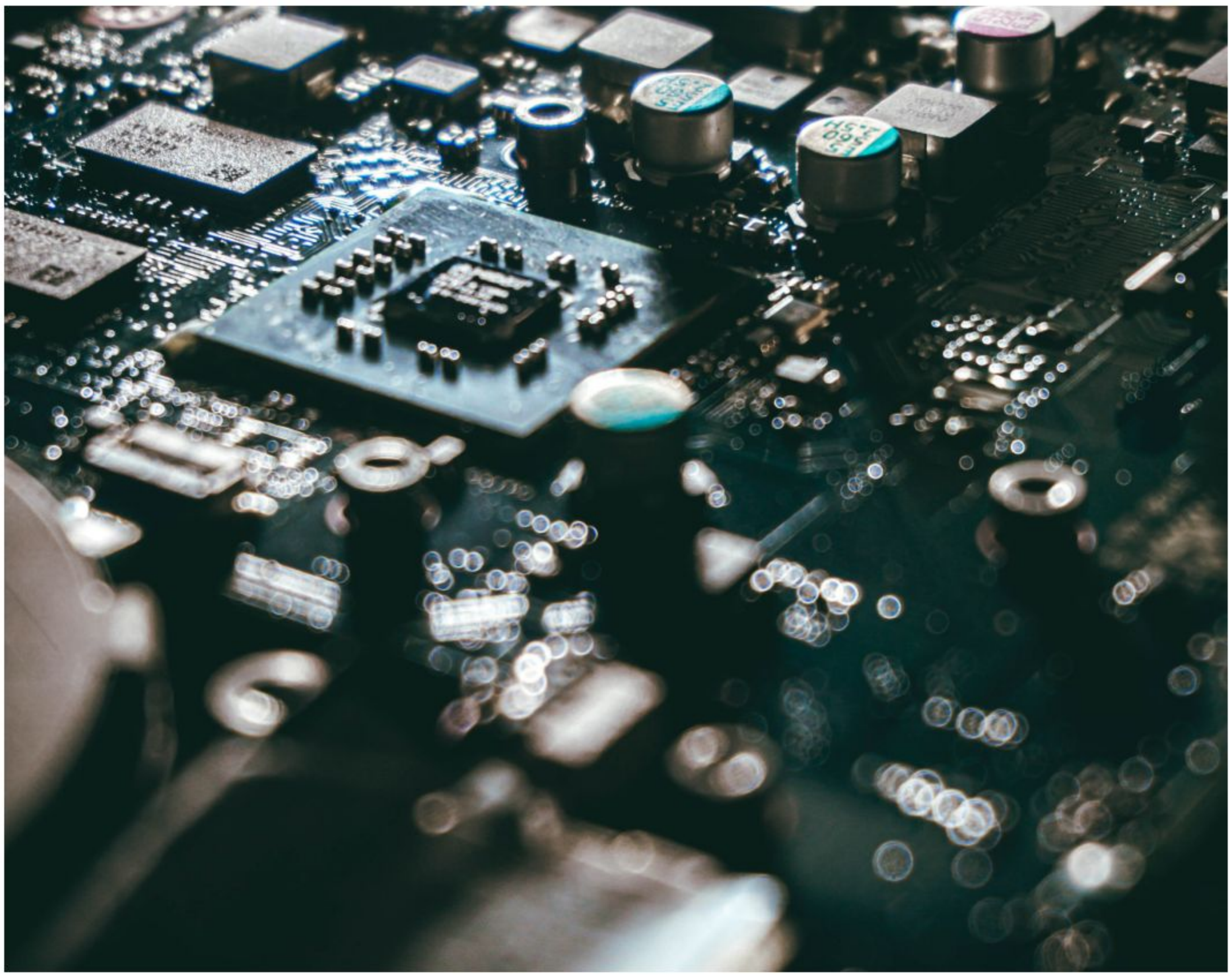
## Electrical Engineering

Provides a broad foundation through combined classroom and laboratory work and prepares students for entering the profession of electrical engineering such as energy production, transport, and distribution, smart grids and design of automated electrical systems. Degree offered: BS in Electrical Engineering.



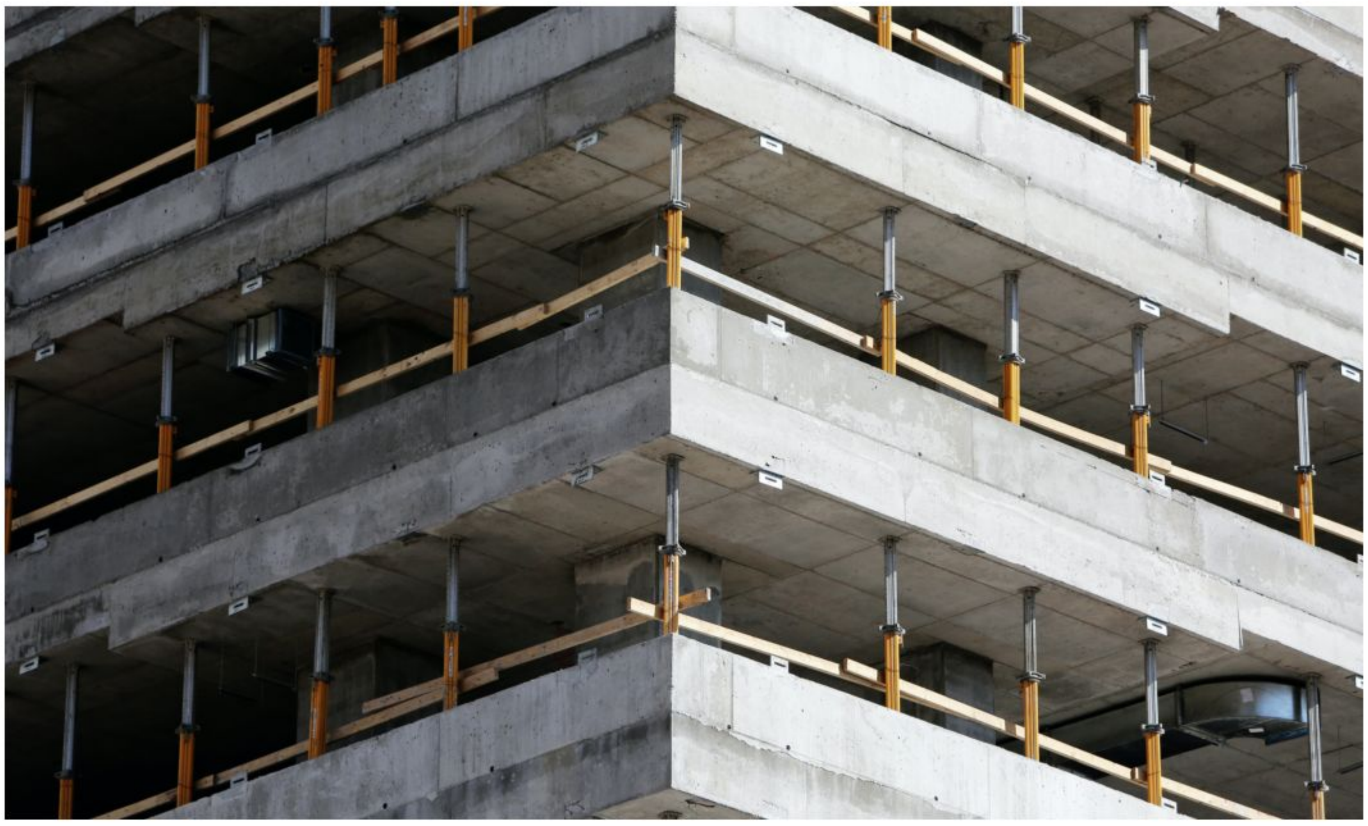
## Computer Engineering

Encompasses the science and technology of design, construction, implementation, testing, and maintenance of integrated software and hardware components of modern computing systems and computer-controlled equipment (cell phones, video games, laptops). Degree offered: BS in Computer Engineering.



## Pre-engineering Program

For students who plan to complete the first two to three years of their engineering education, then either continue at UVU in ABET accredited baccalaureate engineering programs, or transfer to another university. With adequate planning, pre-engineering coursework completed at UVU will be sufficient for students to remain at UVU or to transfer to all the Utah universities with baccalaureate engineering degrees. All students who declare pre-engineering as their major are automatically accepted into preengineering status. Degrees offered: AS, APE (with specific emphasis).

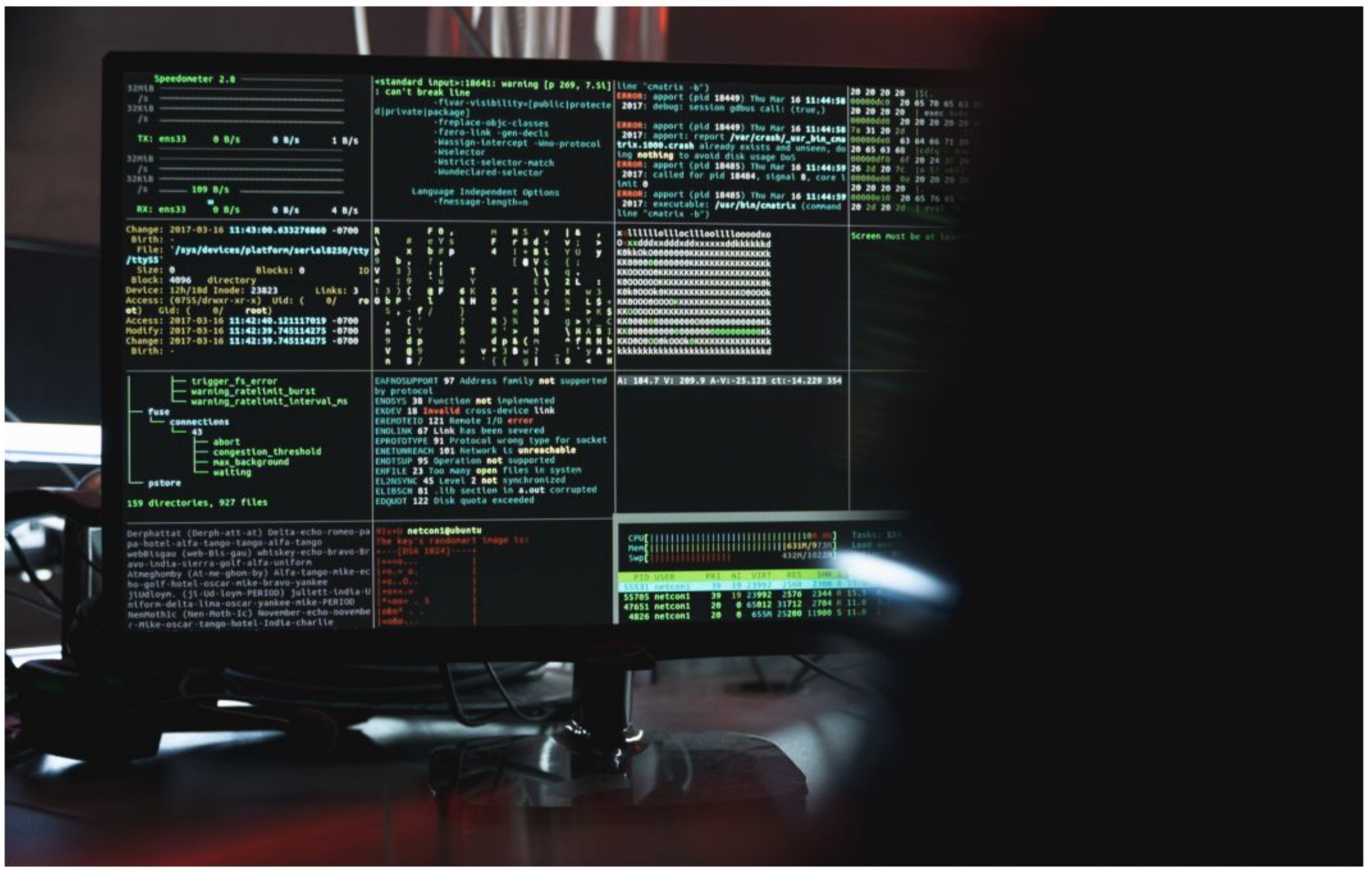



# Information Systems & Technology

# Informtion Systems & Technology

## Information Systems

Involves developing and deploying enterprise-level systems to meet organizational needs. It involves collecting, filtering, creating, processing & interpreting information. IS graduates have a strong business curriculum complimented by courses that focus on utilizing and creating information systems to solve business problems. We offer emphases that focus on Business Intelligence, Application Development, and Information Security Management. Degrees offered: AS, AAS, BS, CP, Minor



## Information Technology

includes installing, managing, and maintaining the computing infrastructure on which organizational systems run. IT prepares students to work as data communication consultants, information security analysts, and network administrators. The core of the IT program prepares students with a strong foundation in Computer Architecture, Data Communication, Cybersecurity, Networks, and System Administration. Degrees offered: AS, AAS, BS, CP, CC

## Information Management

Designed to prepare students to supervise and manage the operations and personnel of business and industry. Information managers are involved in the acquisition of information, the custodianship, and the distribution of that information to those who need it, and its ultimate disposition through archiving or deletion. Degrees offered: AS, AAS, CC, BS

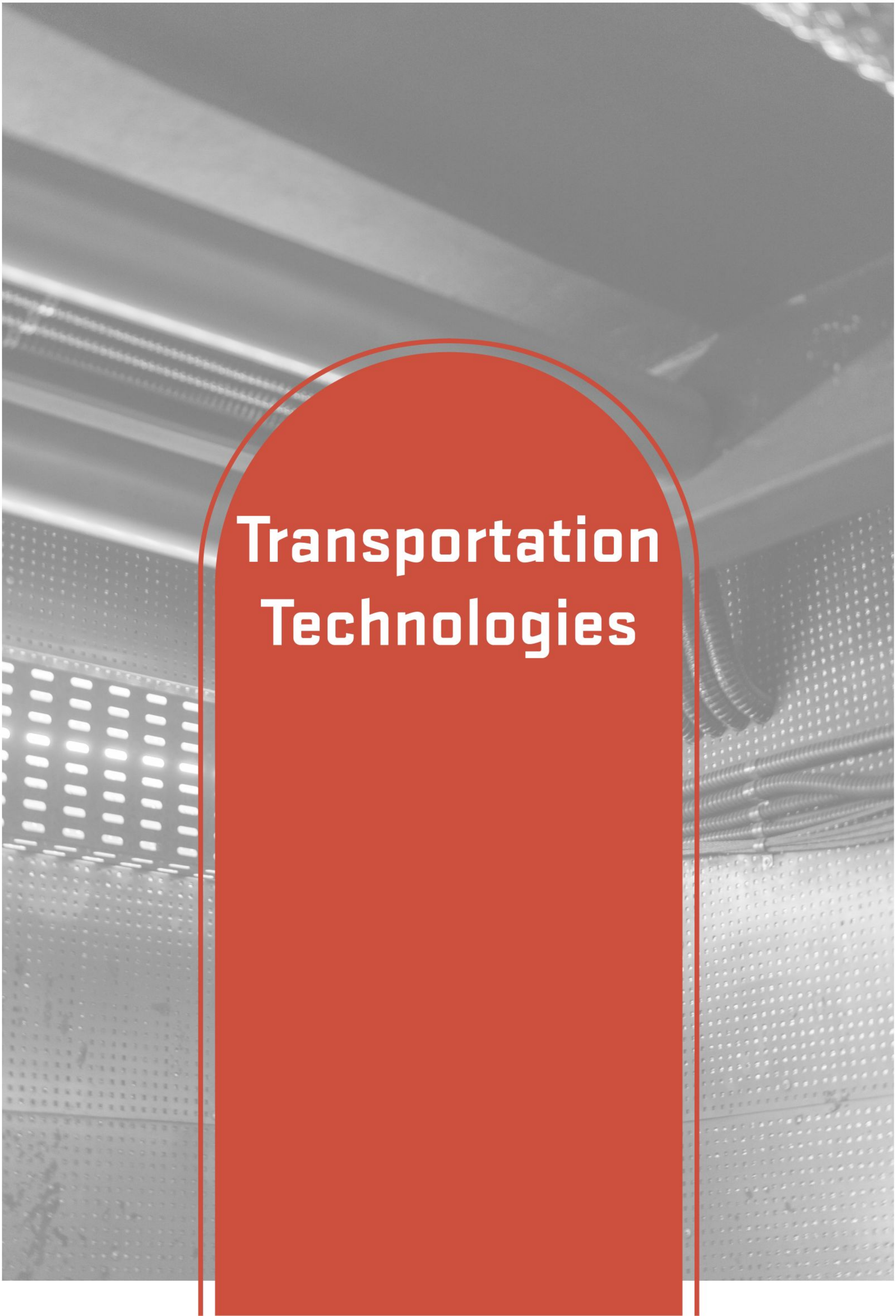
## Business & Marketing Education

Prepares graduates to teach the fundamentals of Business, Marketing, and Business Information Technology to Junior High and High School students in the State of Utah. Degrees offered: BS

## Informtion Systems & Technology

# Master of Science & Cyber Security

Equips technology professionals to safeguard our society and our world from threats such as monetary or identity theft, disruption of defense systems, elections, transportation or energy infrastructure, and other business needs by assessing security needs, creating, managing, and maintaining security systems and protocols. Whether you want to start a new career as a cybersecurity professional, analyst, or specialist, looking to pursue a doctorate or currently working in cybersecurity, this program focuses on the managerial and technical perspectives of cybersecurity through extensive use of case-studies and hands-on lab exercises.



# Transportation Technologies

## Collision Repair

is a two-year AAS Degree program that provides students with the ability to learn industries best practices in Surface Preparation, Nonstructural Repair, Welding, Refinishing, Color Matching, Detailing, Blending, Structural Damage Analysis, Repair and Replacement, Advanced Vehicle Systems diagnostics and repair and Plastic/ Composite Repair. Degrees offered: 1-year certificate, 2-year diploma, AAS.

## Diesel Mechanics

A two-year AAS Degree program that provides students with the ability to learn Diesel engine and diagnostics, hydraulics, electricity/electronics, power train, and chassis. Degrees offered: 1-year certificate, 2-year diploma, AAS.

# Transportation Technologies

## Automotive Technology

A two-year AAS Degree program that provides students the ability to learn on the latest vehicle technologies with the most up-to-date equipment on steering/suspension systems, alignment diagnosis and repair, computer data retrieval, computer updates and programming, electrical diagnosis, automotive propulsion and guidance systems, autonomous vehicle data systems, hybrid and electric vehicle systems, and autonomous driver assisted systems (ADAS). Degrees offered: 1-year certificate, 2-year diploma, AAS.

## Transportation Technologies

# Powersport Technology

A two-year AAS Degree program that provides students with the ability to learn about the many different platforms in the powersports industry. These machines include snowmobiles, personal watercraft, motorcycles, ATV's, UTV's, and many different types of outdoor power equipment. Degrees offered: AAS

# DEGREE OFFERINGS
# PER DEPARTMENT

| Department | AAS | AS | BS | CC | CP | Diploma | GC | Minor | MS |
|---|---|---|---|---|---|---|---|---|---|
| Architecture & Engineering Design | x | x | x | | x | | | | |
| Aviation Science | x | x | x | | | | | | |
| Civil Engineering * | | | x | | | | | | |
| Computer Engineering * | | | x | | | | | | |
| Computer Science | x | x | x | x | | | | x | x |
| Construction Technologies | x | x | x | x | x | x | | | |
| Digital Media | x | | x | | | | | x | |
| Electrical Engineering * | | | x | | | | | | |
| Engineering Technology | x | x | x | | x | | | | |
| Information Systems & Technology | x | x | x | x | x | | x | x | x |
| Mechanical Engineering * | | | x | | | | | | |
| Technology Management | x | | x | | x | | | x | x |
| Transportation Technologies | x | | | x | | x | | | |
| Pre-Engineering | x | x | | | | | | | |

**Key:** AAS: Associate in Applied Science

AS: Associate of Science

BS: Bachelor of Science

CC: Certificate of Completion

CP: Certificate of Profieciency

GC: Graduate Certificate

MS: Master of Science

# Sample Career Data Corresponding to Programs at the UVU College of Engineering & Technology

| CET Program | Median Salary Range (May 2020) | Job Growth Outlook (%) (2019 – 29) |
|---|---|---|
| Architecture & Engineering Design | $46,200 - $149,530 | 1.16% |
| Aviation Science | $31,500 - $134,180 | 4.80% |
| Computer Science | $65,450 - $151,150 | 6.00% |
| Construction Technologies/Construction Management | $29,080 - $97,180 | 1.75% |
| Digital Media | $41,280 - $103,620 | 3.82% |
| Engineering (Mechanical, Electrical, Computer, Civil) | $88,570 - $149,530 | 3.78% |
| Engineering Technology | $37,270 - $89,090 | 3.20% |
| Information Systems & Technology | $34,440 - $185,950 | 12.18% |
| Technology Management | $55,510 - $185,950 | 7.13% |
| Transportation Technologies | $34,170 - $70,200 | 5.50% |

All data obtained from the U.S. Bureau of Labor Statistics (bls.gov)

www.bls.gov

Have any questions or want to learn more? Visit uvu.edu/cet or reach out directly to the center below.

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UTAH VALLEY

UVU

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